

TEAM ASSIGNMENT #3

Course: CHE 433. PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2019
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Due Date: Friday October 4th by 3:00pm in my office

Number of points.....100
Your score.....100.....

Note: All problems are worth 100 points. You final score will be the average value

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1 Executive Summary

1.A The goal of this section was to create a water pipeline simulation with a minimal horsepower value for the pump supplying the flow. In this pipeline wastewater was transported through a series of pipes with two splitters and two mixers. With only one outflow, the pressure from that pipe needed to be matched with atmospheric pressure of 1 atm by changing the pipe power to 520hp. Each splitter in the pipeline needed a split fraction so that pipe pressures going into the mixers were the same and this was done using the ASPEN Plus design specification tool.

1.B In part B, using the ASPEN Plus template: **SteamPipeTemplate.bkp**, steam was being transported through a defined pipe type and length. The optimal steam pipe diameter was needed to be calculated at different pressure drops across the pipe. Using capital cost (CAPEX, 1) and operating cost (OPEX, 2) equations from "A Shortcut to Determine Optimal Steam Pipe Diameter" [1], aspen ran the simulation across the steam pipe at different specified pressure drops and iterate through each desired diameter. The minimum diameter for each specified pressure drop was identified by finding the minimum total cost by adding the CAPEX and OPEX.

1.C Using the ASPEN Plus template: **RefrigerationTemplate.bkp** the goal was to optimize a propane refrigeration cycle. The cycle aimed to remove 3 MW of heat from a process gas stream through the use of a chiller operating at -35 °C. Using the design specification option in ASPEN PLUS the duty of the chiller was set to the desired value of 3 MW with varying the molar flow rate of the propane refrigerant. To further automate the process flowsheet the transfer block in ASPEN PLUS was utilized to update the pressured of the first valve and the flash tank every time the interstage pressure was changed. This was done to ensure that the two streams entering the mixer always had the same pressure. Two more transfer blocks were used in the cycle, first to set the pressure exiting the second compressor to the pressure exiting the condenser, and second, to set the pressure exiting the second valve to the pressure exiting the chiller. This was done because pressure is said to be constant throughout the condenser and chiller. The optimization tool was then used to find the interstage pressure that would give the lowest input power and the higher coefficient of performance(COP).

1.D Using the ASPEN Plus template: **CHProductionTemplate.bkp** the goal was to update a cyclohexane production cycle to reach the production target of 100 KMTA. The design specification option in ASPEN Plus was used to make sure the hydrogen to benzene in the reactor remained at 3.3 by changing the purge fraction of the top splitter (B4). The second design specification was set to make sure the benzene recycle stream flow rate did not go over 45% of the fresh benzene feed flow rate. This was regulated by changing the split fraction of the bottom splitter (B5). The optimal split fraction of the top splitter was 0.04 towards the purge stream and bottom splitter fraction was 0.3 towards the liquid recycle stream. The production target was reached by optimizing the two feeds entering the cyclohexane production cycle.

2 Systems and Procedures

2.1 Components

A. Wastewater Treatment	B. Steam Pipe	C. Refrigeration Cycle	D. Cyclohexane Production
Wastewater	Saturated low-pressure steam	Propane	Benzene Hydrogen Methane Nitrogen Cyclohexane

2.2 Properties

A. STEAMNBS

B. STEAMNBS

C. PENG-ROBINSON

D. RK-SOAVE

2.3 Equipment

A. Pump, two splitters, two mixers, seven pipes

Length of pipes in meters

Q01	Q12	Q13	Q23	Q24	Q34	Q45
1500	300	2000	300	1200	1500	300

B. 100 ft pipe with varying diameters

C. Chiller, condenser, mixer, two valves, two compressors, flash tank

D. Heater, reactor, high-pressure separator, low-pressure separator, two splitters

2.4 Design Specifications

A. Volumetric flowrate of wastewater = 1500 GPM

Temperature of wastewater = 60 °F

All pipes are 6-inch schedule 40 carbon steel

Pump efficiency = 75%

Inlet pressure of feed and outlet pressure of stream 14 = 1 atm

B. Flowrate of saturated steam = 774 SCFM

Average lifetime roughness = 0.5 mm

Boiler pressure range = 1 – 2 bar

Pipe length = 100 ft

C. Chiller duty = 3 MW

Operating temperature of chiller = - 35 °C

Operating temperature of condenser = 35 °C

Compressor 1 and 2 isentropic efficiency = 75%

D. Hydrogen to benzene ratio in the reactor = 3.3

Benzene recycle stream flowrate to Benzene fresh feed flowrate ratio ≤ 0.45

Production target of cyclohexane = 100 KMTA

3 Results

3.A Wastewater is flowing with a total volumetric flow rate of $Q = 1500 \text{ gpm}$ in a pipeline network. The wastewater is at 60°F temperature and atmospheric pressure and it is discharged at the end of the pipeline network in an open pool that is subsequently used for reclamation. All the pipe segments in the network are 6-inch schedule 40 carbon steel. Create an ASPEN Plus simulation to calculate the minimum horsepower requirement of the pump, assuming 75% efficiency.

Minimum pump power required: 520 hp

Split fraction 1: 0.29

Split Fraction 2: 0.26

3.B 774 SCFM of low pressure saturated steam is to be transported in a steam pipeline with an average lifetime roughness of $\varepsilon = 0.5 \text{ mm}$. The boiler will be operating at a pressure range $[1 \text{ bar to } 20 \text{ bar}]$. Develop a model for the calculation of the optimal steam pipe diameter as a function of the boiler pressure.

When observing figure 1 below, the trend for each optimal diameter as pressure changes can be approximated without the need to rerun the data simulation. This graph provides a useful quick relationship between the two when determining the overall desired minimum cost at different decided pressure drops.

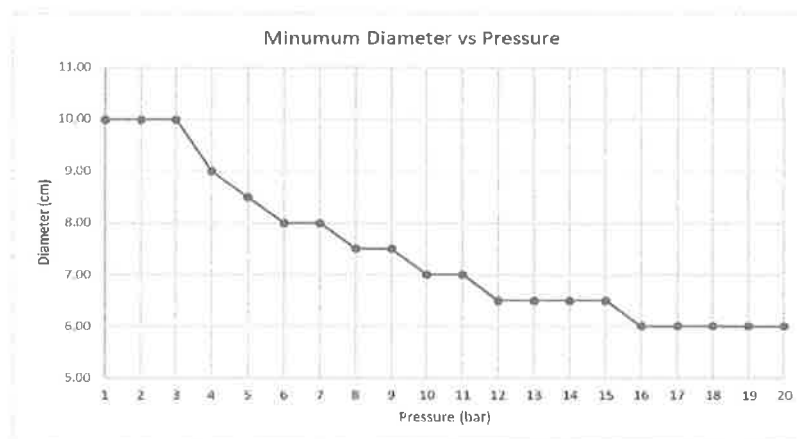


Figure 1: Optimal Diameter vs. Pressure

The table below provides some sample data values between the aspen plus simulated minimum diameter and the optimal diameter determined from theory calculations. The difference between the two is that the paper determined the pressure drop using theoretical relationships while aspen plus pulled the actual calculated pressure drop right out of the simulation itself. For a detailed total cost analysis curve for each diameter and pressure relationship, please refer to figure 2.

Boiler Pressure, bar	Optimal Diameter, mm (from paper)	ASPEN Plus Optimal Diameter	CAPEX (\$/yr)	OPEX (\$/yr)	TOTAL COST (\$/yr)
1	122.6	100	477	1,055	1,532
2	103.2	100	477	271	749
5	79.6	85	374	113	487
8	69.6	75	310	90	400
10	65.3	70	279	85	365
15	58.1	65	250	58	308
20	53.4	60	222	51	272

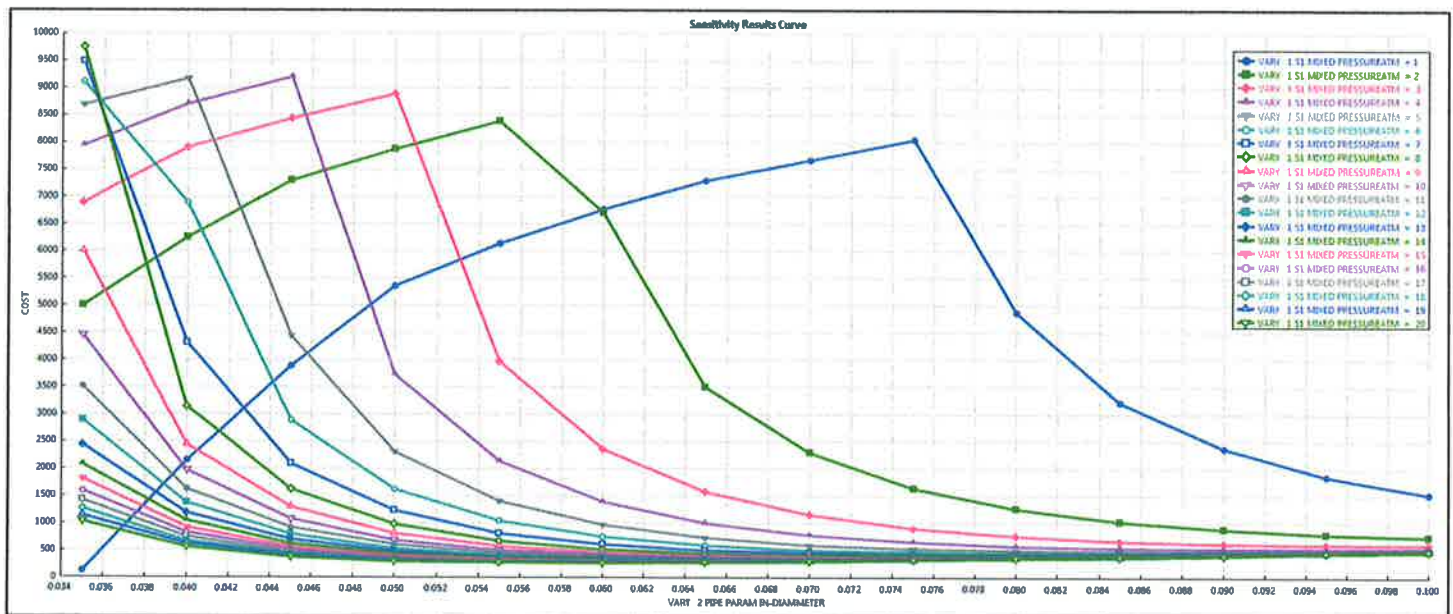


Figure 2: Minimum Total Cost of Operation at Various Pressures

3.C We wish to remove 3 MW from a process gas stream in a chiller operating at $-35\text{ }^{\circ}\text{C}$ and reject it to the environment in a condenser operating at $35\text{ }^{\circ}\text{C}$ using propane as the working fluid.

Conduct a sensitivity analysis and plot the total power requirement as a function of the interstage pressure. Using the Optimization Flowsheeting option determine the optimal interstage pressure for the cycle. What is the COP of the cycle for the optimal pressure and how does it compare with the maximum thermodynamic COP. Assume that both compressors operate at 75% isentropic efficiency.

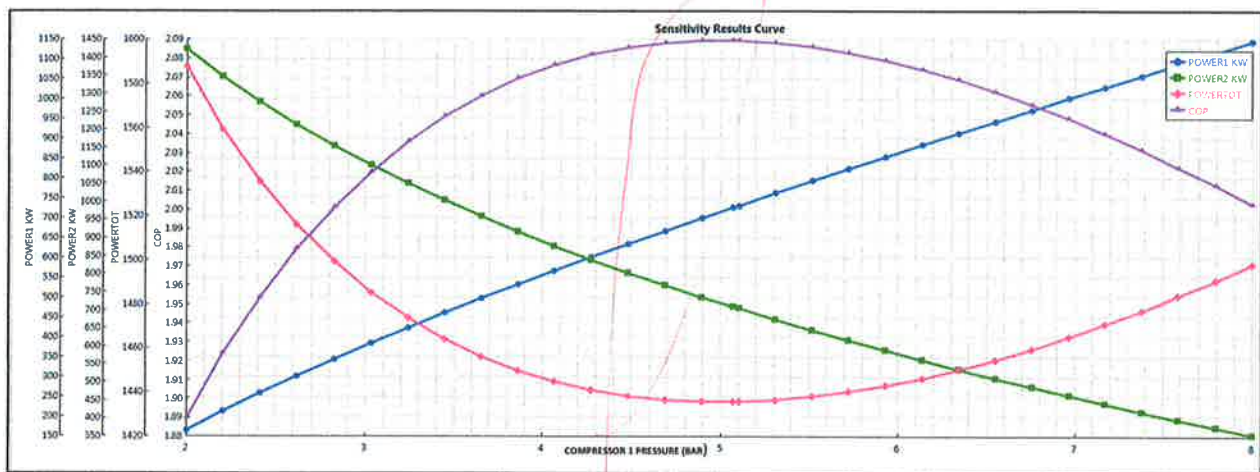


Figure 3: Sensitivity Analysis of Refrigeration Cycle

The optimal interstage pressure for operation was found to be **5.07 bar**, which could be rounded to 5.1 bar for industry purposes. At the interstage pressure of 5.07 bar compressor one required 726.4 kW of power and compressor two required 709.4 kW giving an optimal total power of 1435.8 kW.

The COP of the refrigeration at the optimal pressure was found to be **2.09**. The maximum thermodynamic COP was calculated using equation (3) which gave a value of **3.40**.

3.D For the cyclohexane production process of Assignment #2, prepare design specifications to meet the target ratios for hydrogen and liquid recycle and devise a methodology using the ASPEN Plus Flowsheeting options to calculate the raw material flow rates for a production target of 100 KMTA. Prepare a Custom Table displayed on the flowsheet that would show the results of your methodology.

Streams		BZFED	EFFLUENT FUEL		H2FEED	HPBOT	HPOVHD	PRODUCT PURGE		RECGAS	RECLIQ	TEAR	TOPURIFY
From			REACTOR LPSEP			HPSEP	HPSEP	LPSEP	B4	B4	B5	HEATER	B5
To		HEATER	HPSEP		HEATER	B5	B4			HEATER	HEATER	REACTOR	LPSEP
Phase:		Liquid	Vapor	Vapor	Vapor	Liquid	Vapor	Liquid	Vapor	Vapor	Liquid	Mixed	Liquid
Component Mole Flow													
H2	LBMOL/HR	0	153.0912	1.925375	947.7443	2.772139	150.319	0.028574	30.06381	120.2552	0.8181893	1068.818	1.953949
N2	LBMOL/HR	0	6.927014	0.2336474	1.56	0.3457502	6.581264	0.0100555	1.316253	5.265011	0.1020472	6.927014	0.243703
C1	LBMOL/HR	0	44.62927	3.482364	6.24	5.52043	39.10884	0.4087258	7.821767	31.28707	1.62934	39.1568	3.89109
BZ	LBMOL/HR	303.8439	0.6080521	0.0039917	0	0.6027593	0.0052928	0.4208649	0.0010586	0.0042343	0.1779028	323.636	0.4248565
CH	LBMOL/HR	0	431.7099	2.654343	0	428.1709	3.53903	299.1431	0.7078061	2.831224	126.3735	129.204	301.7974
Component Mole Fraction													
H2		0	0.2403446	0.2319807	0.9918371	0.0063376	0.753277	9.52E-05	0.753277	0.753277	0.0063376	0.690392	0.0063376
N2		0	0.010875	0.0281512	0.0016326	0.0007904	0.0329799	3.35E-05	0.0329799	0.0329799	0.0007904	0.0044744	0.0007904
C1		0	0.0700654	0.419576	0.0065303	0.0126206	0.1959818	0.0013624	0.1959818	0.1959818	0.0126206	0.0252929	0.0126206
BZ		1	0.0009546	0.0004809	0	0.001378	2.65E-05	0.0014028	2.65E-05	2.65E-05	0.001378	0.1963826	0.001378
CH		0	0.6777604	0.3198111	0	0.9788733	0.0177347	0.997106	0.0177347	0.0177347	0.9788733	0.083458	0.9788733
Mole Flow	LBMOL/HR	303.8439	636.9654	8.299721	955.5443	437.412	199.5535	300.0113	39.91069	159.6428	129.101	1548.132	308.311
Mass Flow	LB/HR	23734.35	37599.4	289.998	2054.347	36186.33	1413.066	25216.03	282.6131	1130.453	10680.3	37599.4	25506.03
Temperature	F	100	400	118.521	120	120	120	118.521	120	120	120	300	120
Pressure	PSIA	15	315	15	335	310	310	15	310	310	310	330	310
Average Molecular Weight		78.11364	59.02894	34.94069	2.149923	82.72826	7.081139	84.05029	7.081139	7.081139	82.72826	24.28695	82.72826

Figure 3: Cyclohexane Production

The custom table shows that the specified ratios have been maintained. The stream leaving the heater and entering the reactor (outlined in red) has a hydrogen to benzene ratio of 3.3 and the recycle benzene to fresh feed ratio has a ratio of 0.45. The product flowrate (outlined in green) is shown to be 25216.03-pound moles and hour which converted to reach a 100 KMTA of cyclohexane.

4 Conclusion and Discussion

4.A Water flow through a pipeline needs to be carefully directed so the splitters aren't random values that need to be declared. Splitter 1's fraction value of 0.29 makes sure that the pipes going into the mixer 1 are both at 9.47atm while splitter 2's fraction value of 0.26 makes sure that the pipes going into mixer 2 are both at 4.96atm.

Wastewater treatment and pipeline design are commonly found in industry such as the water reclamation service where water transport is their profession. While this PFD concerned itself with only one feed and one outlet. Real world scenarios find themselves with multiple feeds of wastewater coming from different sources and they would all cumulate together at one outlet. This diagram stimulates that scenario of pipes interconnecting and bringing together different sources of wastewater except the multitude of feed sources was simplified to one feed.

4.B From analyzing the data in the tables seen in section **3B**, as the pressure drop increased, the diameter decreased to accommodate for a lower total cost. An x-y relationship made from the diameter as a function of pressure drop is found in figure 1. Using the relationship graph, intermediary pressure drops, and their resulting minimum diameters can be approximated without the need to run further aspen simulations. The global minimum for the total cost comparative to diameter and pressure is found at 20 bar and 60mm.

Optimization is one of the most important techniques in industry since it can save companies thousands if not millions of dollars in production per year. In this simple steam pipe example, the expenditure for steam running through a pipe could have ranged anywhere from \$272/yr to \$1,532/yr. The same process is occurring but just at different defined steam and pipe properties. Small changes in the process such as diameter or length and flowrate or pressure can save companies millions and by being skilled in knowing what values to optimize, one can spend multitude less cost and earn back more, making the companies footing and value in industry bigger than its competitors.

4.C The coefficient of performance found in the propane refrigeration cycle was within the typical range of a refrigeration cycle, which can usually be found to be between 2 – 4 [2]. The value of the actual COP is lower than the calculated thermodynamic maximum which is expected and further confirms that it is reasonable. If the COP had been lower than the acceptable range, the refrigeration cycle would be deemed inefficient and a different refrigerant would have to be considered.

4.D The cyclohexane production cycle was required to produce a 100 KMTA of product and this was reached by optimizing the flowrates of the benzene and hydrogen feeds into the system. The specific purge fractions were found using ASPEN Plus to meet the design specifications given in the problem.

5 Appendices**5.1 Equations**

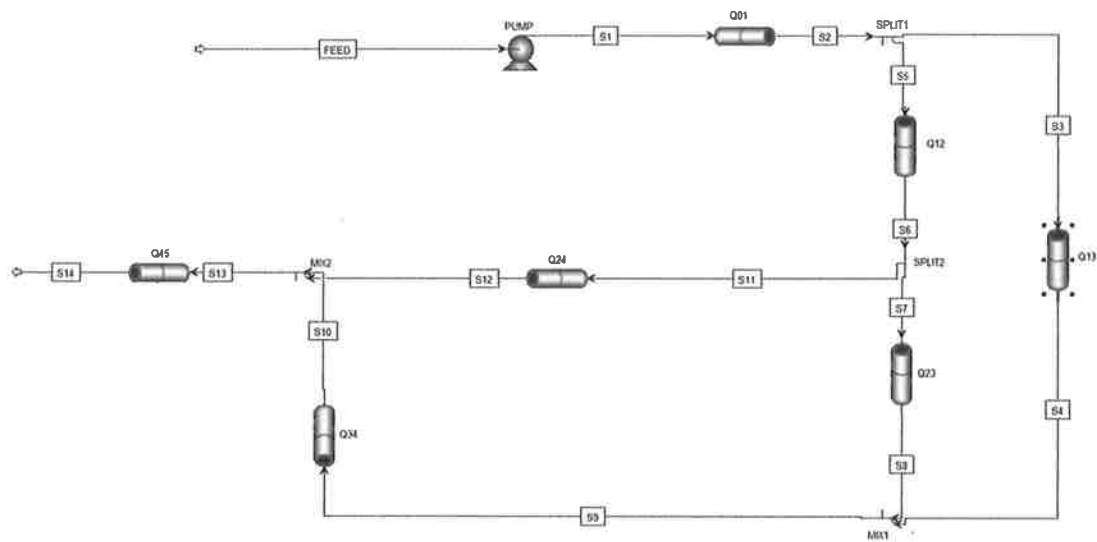
$$CAPEX = mD^n L(1 + F)(a + b) \quad (1)$$

$$OPEX = A(\Delta P) \left(\frac{G}{\rho} \right) (a + b)(1 + F)(mn) \quad (2)$$

$$Thermodynamic \ Max \ COP = \frac{T_c}{T_h - T_c} \quad (3)$$

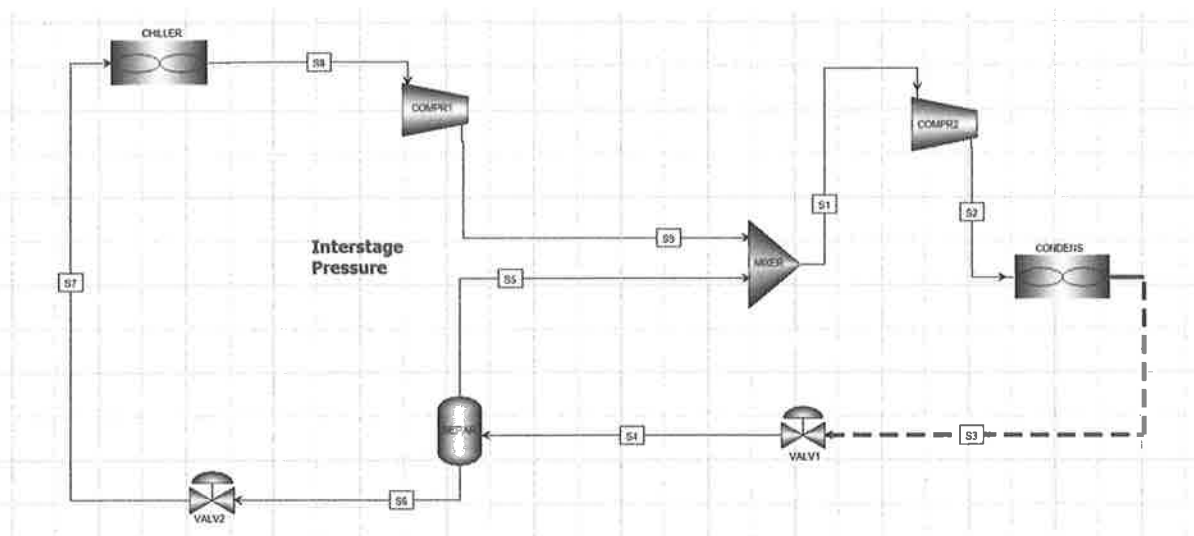
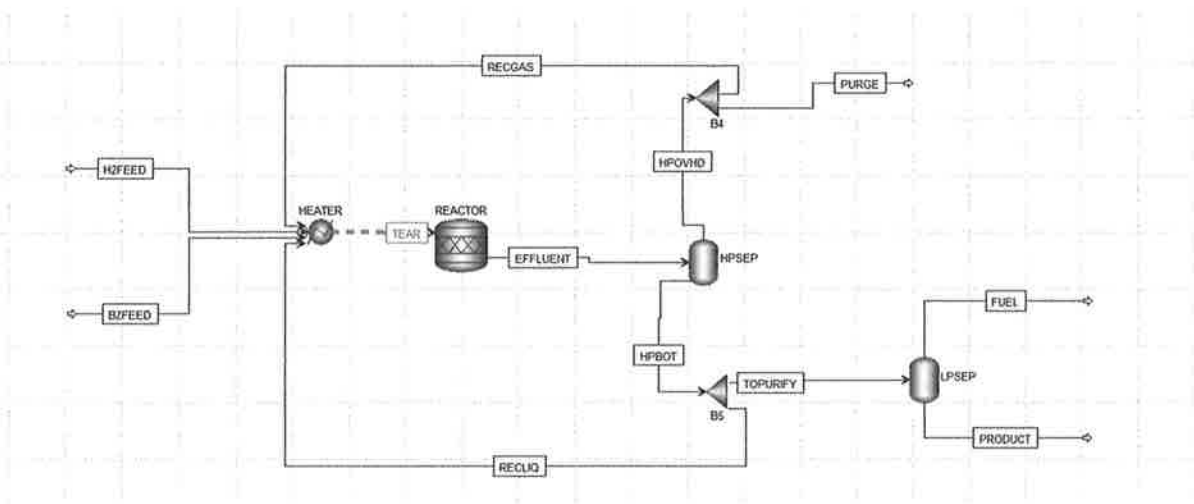
5.2 PFD

A. Wastewater Treatment



B. Optimal Diameter of a Steam Pipe



C. Refrigeration with Flask Tank Economizer and Two Stages of Compression**D. Cyclohexane Production**

5.3 Mass and Energy Balances

Part A: Wastewater Treatment:

Pump			
	FEED	S1	
Total Flow kg/sec	94.5	94.5	
Temperature K	289	289	
Pressure N/sqm	1.01E+05	3.18E+06	
Enthalpy J/kg	-1.59E+07	-1.59E+07	
Entropy J/kg-K	-9.19E+03	-9.19E+03	
Pipe - Q01			
	S1	S2	
Total Flow kg/sec	94.5	94.5	
Temperature K	289	289	
Pressure N/sqm	3.18E+06	1.19E+06	
Enthalpy J/kg	-1.59E+07	-1.59E+07	
Entropy J/kg-K	-9.19E+03	-9.18E+03	
Splitter 1			
	S2	S3	S5
Total Flow kg/sec	94.5	26.3	68.3
Temperature K	289	289	289
Pressure N/sqm	1.19E+06	1.19E+06	1.19E+06
Enthalpy J/kg	-1.59E+07	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03	-9.18E+03
Pipe - Q12			
	S5	S6	
Total Flow kg/sec	68.3	68.3	
Temperature K	289	289	
Pressure N/sqm	1.19E+06	9.77E+05	
Enthalpy J/kg	-1.59E+07	-1.59E+07	
Entropy J/kg-K	-9.18E+03	-9.18E+03	
Splitter 2			
	S6	S7	S8
Total Flow kg/sec	68.28059	17.81531	17.81531
Temperature K	289.4886	289.4886	289.4924
Pressure N/sqm	976642	976642	960054
Enthalpy J/kg	-1.6E+07	-1.6E+07	-1.6E+07
Entropy J/kg-K	-9179.62	-9179.62	-9179.56

Pipe - Q13

	S3	S4
Total Flow kg/sec	26.3	26.3
Temperature K	289	289
Pressure N/sqm	1.19E+06	9.60E+05
Enthalpy J/kg	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03

Pipe - Q23

	S7	S8
Total Flow kg/sec	17.8	17.8
Temperature K	289	289
Pressure N/sqm	9.77E+05	9.60E+05
Enthalpy J/kg	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03

Mixer 1

	S4	S8	S9
Total Flow kg/sec	26.3	17.8	44.1
Temperature K	289	289	289
Pressure N/sqm	9.60E+05	9.60E+05	9.60E+05
Enthalpy J/kg	-1.59E+07	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03	-9.18E+03

Pipe - Q24

	S11	S12
Total Flow kg/sec	50.5	50.5
Temperature K	289	290
Pressure N/sqm	9.77E+05	5.03E+05
Enthalpy J/kg	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03

Pipe - Q34

	S9	S10
Total Flow kg/sec	44.1	44.1
Temperature K	289	290
Pressure N/sqm	9.60E+05	5.03E+05
Enthalpy J/kg	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03

Mixer 2			
	S10	S12	S13
Total Flow kg/sec	44.1	50.5	94.5
Temperature K	290	290	290
Pressure N/sqm	5.03E+05	5.03E+05	5.03E+05
Enthalpy J/kg	-1.59E+07	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03	-9.18E+03

Pipe - Q45		
	S13	S14
Total Flow kg/sec	94.5	94.5
Temperature K	290	290
Pressure N/sqm	5.03E+05	1.04E+05
Enthalpy J/kg	-1.59E+07	-1.59E+07
Entropy J/kg-K	-9.18E+03	-9.18E+03

Part C: Refrigeration with Flash Tank Economizer and Two Stages of Compression

Chiller			
	S7	S8	
Total Flow kg/hr	32936.48	32936.48	
Temperature C	-34.9391	-35	
Pressure bar	1.38	1.376565	
Vapor Frac	0.213417	1	
Liquid Frac	0.786583	0	
Enthalpy kJ/kg	-2800.44	-2472.53	
Entropy kJ/kg-K	-7.90466	-6.52769	
Density gm/cc	0.014729	0.003201	
Compressor 1			
	S8	S9	
Total Flow kg/hr	32936.48	32936.48	
Temperature C	-35	21.79866	
Pressure bar	1.376565	5.07	
Vapor Frac	1	1	
Liquid Frac	0	0	
Enthalpy kJ/kg	-2472.53	-2393.13	
Entropy kJ/kg-K	-6.52769	-6.45904	
Density gm/cc	0.003201	0.010033	
Mixer			
	S1	S5	S9
Total Flow kg/hr	43306.5	10370.02	32936.48
Temperature C	17.16798	2.220644	21.79866
Pressure bar	5.07	5.07	5.07
Vapor Frac	1	1	1
Liquid Frac	0	0	0
Enthalpy kJ/kg	-2401.15	-2426.6	-2393.13
Entropy kJ/kg-K	-6.48643	-6.57644	-6.45904
Density gm/cc	0.010242	0.010998	0.010033
Compressor 2			
	S1	S2	
Total Flow kg/hr	43306.5	43306.5	
Temperature C	17.16798	60.24208	
Pressure bar	5.07	12.21	
Vapor Frac	1	1	
Liquid Frac	0	0	
Enthalpy kJ/kg	-2401.15	-2342.18	
Entropy kJ/kg-K	-6.48643	-6.44171	
Density gm/cc	0.010242	0.023214	

Condenser		
	S2	S3
Total Flow kg/hr	43306.5	43306.5
Temperature C	60.24208	35
Pressure bar	12.21	12.20618
Vapor Frac	1	0
Liquid Frac	0	1
Enthalpy kJ/kg	-2342.18	-2710.92
Entropy kJ/kg-K	-6.44171	-7.63197
Density gm/cc	0.023214	0.475955

Valve 1		
	S3	S4
Total Flow kg/hr	43306.5	43306.5
Temperature C	35	2.220644
Pressure bar	12.20618	5.07
Vapor Frac	0	0.239456
Liquid Frac	1	0.760544
Enthalpy kJ/kg	-2710.92	-2710.92
Entropy kJ/kg-K	-7.63197	-7.60893
Density gm/cc	0.475955	0.043068

Separator			
	S4	S5	S6
Total Flow kg/hr	43306.5	10370.02	32936.48
Temperature C	2.220644	2.220644	2.220644
Pressure bar	5.07	5.07	5.07
Vapor Frac	0.239456	1	0
Liquid Frac	0.760544	0	1
Enthalpy kJ/kg	-2710.92	-2426.6	-2800.44
Entropy kJ/kg-K	-7.60893	-6.57644	-7.93401
Density gm/cc	0.043068	0.010998	0.525893

Valve 2		
	S6	S7
Total Flow kg/hr	32936.48	32936.48
Temperature C	2.220644	-34.9391
Pressure bar	5.07	1.38
Vapor Frac	0	0.213417
Liquid Frac	1	0.786583
Enthalpy kJ/kg	-2800.44	-2800.44
Entropy kJ/kg-K	-7.93401	-7.90466
Density gm/cc	0.525893	0.014729

Part D: Cyclohexane Production

	Heater			
	BZFEED	H2FEED	RECGAS	RECLIQ
Mole Flow lbmol/hr				
H2	0	304.2	25.88536	0.1141378
N2	0	1.56	12.20695	0.1499972
C1	0	6.24	45.6519	1.479822
BZ	100	0	0.00252102	0.0580085
CH	0	0	1.709188	41.37693
Mole Frac				
H2	0	0.975	0.3029089	0.00264337
N2	0	0.005	0.142845	0.00347386
C1	0	0.02	0.5342157	0.0342718
BZ	1	0	2.95E-05	0.00134345
CH	0	0	0.0200008	0.9582674
Total Flow lbmol/hr	100	312	85.45593	43.17889
Temperature F	100	120	120	120
Pressure psia	15	335	310	310
Vapor Frac	0	1	1	0
Liquid Frac	1	0	0	1
Enthalpy Btu/lbmol	21902.01	-334.2858	-17966.39	-63870.87
Entropy Btu/lbmol-R	-58.89429	-5.809383	-16.21523	-138.655
Density lbmol/cuft	0.6866711	0.0531551	0.0500494	0.5668414

Reactor			
	EFFLUENT	TEAR	
Mole Flow lbmol/hr			
H2	28.81743		330.1997
N2	13.91684		13.91684
C1	55.17265		53.37156
BZ	0.2001211		100.0605
CH	142.6466		43.08633
Mole Frac			
H2	0.1196968		0.6107628
N2	0.0578053		0.0257416
C1	0.2291665		0.0987201
BZ	0.000831228		0.1850796
CH	0.5925002		0.0796957
Total Flow lbmol/hr	240.7536		540.635
Temperature F	400		300
Pressure psia	315		330
Vapor Frac	1		1
Liquid Frac	0		0
Enthalpy Btu/lbmol	-31755.95		1808.391
Entropy Btu/lbmol-R	-71.64606		-18.55062
Density lbmol/cuft	0.0384298		0.0410129
High Pressure Separator			
	EFFLUENT	HPBOT	HPOVHD
Mole Flow lbmol/hr			
H2	28.81743	0.3883108	28.42912
N2	13.91684	0.510309	13.40653
C1	55.17265	5.034538	50.13811
BZ	0.2001211	0.1973523	0.00276876
CH	142.6466	140.7694	1.87715
Mole Frac			
H2	0.1196968	0.00264337	0.3029089
N2	0.0578053	0.00347386	0.142845
C1	0.2291665	0.0342718	0.5342157
BZ	0.000831228	0.00134345	2.95E-05
CH	0.5925002	0.9582674	0.0200008
Total Flow lbmol/hr	240.7536	146.8999	93.85368
Temperature F	400	120	120
Pressure psia	315	310	310
Vapor Frac	1	0	1
Liquid Frac	0	1	0
Enthalpy Btu/lbmol	-31755.95	-63870.87	-17966.39
Entropy Btu/lbmol-R	-71.64606	-138.655	-16.21523
Density lbmol/cuft	0.0384298	0.5668414	0.0500494

	Top Splitter		
	HPOVHD	PURGE	RECGAS
Mole Flow lbmol/hr			
H2	28.42912	2.543756	25.88536
N2	13.40653	1.199578	12.20695
C1	50.13811	4.486214	45.6519
BZ	0.00276876	0.000247741	0.00252102
CH	1.87715	0.167962	1.709188
Mole Frac			
H2	0.3029089	0.3029089	0.3029089
N2	0.142845	0.142845	0.142845
C1	0.5342157	0.5342157	0.5342157
BZ	2.95E-05	2.95E-05	2.95E-05
CH	0.0200008	0.0200008	0.0200008
Total Flow lbmol/hr	93.85368	8.397758	85.45593
Temperature F	120	120	120
Pressure psia	310	310	310
Vapor Frac	1	1	1
Liquid Frac	0	0	0
Enthalpy Btu/lbmol	-17966.39	-17966.39	-17966.39
Entropy Btu/lbmol-R	-16.21523	-16.21523	-16.21523
Density lbmol/cuft	0.0500494	0.0500494	0.0500494

Bottom Splitter			
	HPBOT	RECLIQ	TOPURIFY
Mole Flow lbmol/hr			
H2	0.3883108	0.1141378	0.274173
N2	0.510309	0.1499972	0.3603118
C1	5.034538	1.479822	3.554716
BZ	0.1973523	0.0580085	0.1393437
CH	140.7694	41.37693	99.39248
Mole Frac			
H2	0.00264337	0.00264337	0.00264337
N2	0.00347386	0.00347386	0.00347386
C1	0.0342718	0.0342718	0.0342718
BZ	0.00134345	0.00134345	0.00134345
CH	0.9582674	0.9582674	0.9582674
Total Flow lbmol/hr	146.8999	43.17889	103.721
Temperature F	120	120	120
Pressure psia	310	310	310
Vapor Frac	0	0	0
Liquid Frac	1	1	1
Enthalpy Btu/lbmol	-63870.87	-63870.87	-63870.87
Entropy Btu/lbmol-R	-138.655	-138.655	-138.655
Density lbmol/cuft	0.5668414	0.5668414	0.5668414
Low Pressure Separator			
	FUEL	PRODUCT	TOPURIFY
Mole Flow lbmol/hr			
H2	0.2722606	0.00191242	0.274173
N2	0.3530399	0.00727192	0.3603118
C1	3.364086	0.1906298	3.554716
BZ	0.00250334	0.1368404	0.1393437
CH	1.673979	97.7185	99.39248
Mole Frac			
H2	0.0480527	1.95E-05	0.00264337
N2	0.0623099	7.42E-05	0.00347386
C1	0.5937458	0.00194411	0.0342718
BZ	0.000441828	0.00139555	0.00134345
CH	0.2954496	0.9965667	0.9582674
Total Flow lbmol/hr	5.665869	98.05515	103.721
Temperature F	114.681	114.681	120
Pressure psia	15	15	310
Vapor Frac	1	0	0
Liquid Frac	0	1	1
Enthalpy Btu/lbmol	-34264.79	-65581.58	-63870.87
Entropy Btu/lbmol-R	-45.36174	-143.6729	-138.655
Density lbmol/cuft	0.00245119	0.5586003	0.5668414

5.4 References

- [1] Genic, Srbislav and Jacimovic, Branislav. A Shortcut to Determine Optimal Steam Pipe Diameter. Univ. of Beldgrade, Serbia, 2018, Published: American Institute of Chemical Engineers
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